Technical report TR1: Reproducing OPM Flow regression references (moving reference histories and a well reference-depth subtlety)

Draft v1.0, 2026-11-04. Intended for Zenodo (technical report) and as issues/notes to the OPM project. Not a journal paper. Authors: [to be decided]. Generative-AI assistance was used for code, analysis and drafting; AI tools are not authors.

Summary

Independent simulators are often validated against the reference summary files distributed with the OPM Flow regression suite (opm-tests). We compared an independent black-oil engine (ResSim) with those references over 209 cases. Two observations matter to anyone doing the same:

1. Several reference histories are **moving targets**: OPM's own references for the same deck differ between repository versions by up to 53 %.
2. The default **bottom-hole reference depth** of a well in OPM is taken from its first connection, *including connections defined SHUT*. A simulator that uses the shallowest open connection instead will mismatch BHP-controlled wells by several per cent.

1. Method

- Cases: all 209 decks fetched by `tools/fetch_opm_tests.py` (opm-tests default folders).
- Metric: worst relative RMS over the compared summary vectors, after time alignment; pass at 5 %. A "material" variant ignores vectors below 1 % of their peak. The metric was never loosened during the work.
- Reference drift: the reference SMSPEC/UNSMRY of each case was taken at four opm-tests commits (2021-01 `66a7fbe991`, 2023-04 `d5a01f9c5f`, 2024-09 `67db106e68`, 2025-05 `da782d48cd`) and compared with each other using the same metric.
- Final status of the engine (C49): 102 PASS / 77 FAIL / 30 SKIP. SKIP means unsupported features (e.g. binary IMPORT grids, gas-water systems without oil). Full list: `docs/regression_209_full.txt`.

2. Moving reference histories (F29)

Group	Cases	Spread between OPM's own reference versions
stable	6_ENDSCALE, 9_1A, 0A1, 0B_WELPI, ACTIONX_M1, ACTIONX_WEFAC, 7_HYSTERESIS, WSEGSICD, WSEGAICD	≤ 2.5 %
moderate	9_3A	12–16 %
unstable	NETWORK-01-WTEST, T1L, UDQ_WCONPROD, UDQ_ACTIONX	35–53 %

Bisection of two cases.

- T1L changed with opm-simulators PR 6311.
- 9_3A changed with opm-common PR 3779, which pads low-pressure gas PVT tables (`fvf_lim = max(10·B0, 1)`). The change adds table rows only below 3 bar, yet the reference FWCT differs by 45 % at t = 151 d (31 vs 26 ministeps). This points to transient instability of the case rather than a physics change. ResSim reproduces the 2024-09 reference to 5.4 %.

Recommendation. Validation studies should state the opm-tests commit of their references. Cases in the "unstable" group should not be used as pass/fail criteria without an inter-version tolerance.

3. Well reference depth includes SHUT connections (F34)

- **OPM behaviour** (opm-common `Well::updateRefDepth`): when WELSPECS item 5 is defaulted, the reference depth is taken once from connection 0, whatever its state. (We checked OPM; we did not check

the ECLIPSE manual wording for this edge case.)

- **Symptom** in ACTIONX_WEFAC / GCONINJE: well OP02 has K3–4 defined SHUT, and K7–10 are opened later by WELOPEN on COMPLUMP 2. Using the shallowest *open* connection made WBHP 3 bar (1.2 %) high, so the 200-bar limit was reached 180 d late. Late-time WOPR/WGPR were 15–20 % off.
- **After matching OPM's rule:** WEFAC 9.46 % → 1.8 % (PASS); GCONINJE 10.3 % → 1.04 % (PASS).
- **Related:** GCONPROD/GCONINJE rate items may be UDQ names (group UDA). Supporting this brought ACTIONX_GCONPROD FGPR to 0.00 % up to 1066 d. The remaining 19 % after an ACT-03 control jump comes from the deck itself: a compounding UDA (FU_FGPR = 0.9·FGPR) depends on time-step size.

4. Reproducibility

Scripts: `tools/fetch_opm_tests.py`, `tools/dev/refdrift*`, `tools/dev/smrydiff.py`, `tools/dev/wdump.cpp`, `tools/dev/cmpw.py`. Engine notes: ENGINES_AND_MATH.md §9.t. The opm-tests data are GPL-3.0 / ODbL as distributed by OPM. Only commit hashes and derived statistics are published here.

5. Next step (human)

Open a short issue or discussion in OPM/opm-tests on the inter-version spread of the four unstable cases. Link this report and offer the comparison script.